

FinKont2: Hand-In Exercise #1

Answers (individually composed – but feel free to discuss in groups) must be handed in via Absalon no later than 23:59 CET on Monday February 23, 2026. There are 10 equal-weighted (number.letter)-named questions.

Moments Etc. (30%)

1.a

Consider the Vasicek (or Ornstein/Uhlenbeck) stochastic differential equation (SDE)

$$dX(t) = \kappa(\theta - X(t))dt + \sigma dW(t).$$

Show that

$$X(t+u)|\mathcal{F}_t \sim N\left(X(t)e^{-\kappa u} + \theta(1 - e^{-\kappa u}), \frac{\sigma^2(1 - e^{-2\kappa u})}{2\kappa}\right)$$

Hint: Look at the dynamics of the process defined by $Z(t) = e^{\kappa t}X(t)$ and use Björk's Lemma 4.18.

1.b

Consider the Cox/Ingersoll/Ross (or $\sqrt{\cdot}$) SDE

$$dX(t) = \kappa(\theta - X(t))dt + \sigma\sqrt{X(t)}dW(t), \quad X(0) = x.$$

Put $m(t) = \mathbf{E}(X(t))$. **Show that**

$$m(t) = X(0)e^{-\kappa t} + \theta(1 - e^{-\kappa t}).$$

Hint: Write the SDE on integral form, take mean, interchange (allowed in FinKont2 ;)), and observe that the m -function solves an ordinary differential equation (ODE).

1.c – still considering the Cox/Ingersoll/Ross (or $\sqrt{\cdot}$) SDE

Derive an ODE for the second moment, $h(t) := \mathbf{E}(X^2(t))$. Hint: Use Ito on X^2 and do as in 1.b.

Prove (e.g. by direct verification) **that**

$$\text{var}(X(t)) = X(0)\frac{\sigma^2}{\kappa}(e^{-\kappa t} - e^{-2\kappa t}) + \theta\frac{\sigma^2}{2\kappa}(1 - e^{-\kappa t})^2.$$

Remark: Because X is a time-homogeneous Markov process conditional moments are found by shifting the formulas/equations.

The Bachelier Model (20%)

The Bachelier model – named after the French finance (and stochastic processes) pioneer Louis Bachelier who used it in his 1900 dissertation – models the price of a stock, S , by a so-called arithmetic (as opposed to geometric) process, which has dynamics of the form

$$dS(t) = \dots dt + \sigma dW(t),$$

the important point being that the term in front of $dW(t)$ is constant, i.e. it does not have $S(t)$ itself in it.

2.a

Suppose the interest rate is 0 and consider a strike- K expiry- T call-option. **Show that** its arbitrage-free time- t price is

$$\pi^{\text{call, Bach}}(t) = (S(t) - K)\Phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right) + \sigma\sqrt{T-t}\phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right),$$

where Φ and ϕ denote, respectively, the standard normal distribution and density function. **Show that** the time- t Δ -hedge-ratio of the call-option is

$$\Delta^{\text{call, Bach}}(t) = \Phi\left(\frac{S(t) - K}{\sigma\sqrt{T-t}}\right).$$

Hint: You may without proof use that if $X \sim N(\mu, \sigma^2)$ then

$$\mathbb{E}(X\mathbf{1}_{l \leq X \leq h}) = \mu\left(\Phi\left(\frac{h-\mu}{\sigma}\right) - \Phi\left(\frac{l-\mu}{\sigma}\right)\right) + \sigma\left(\phi\left(\frac{l-\mu}{\sigma}\right) - \phi\left(\frac{h-\mu}{\sigma}\right)\right),$$

where as usual $\mathbf{1}$ denotes the indicator function.

What does the call-price formula look like in a Bachelier model with a non-zero (but constant) interest rate r ? Hint: Use the trick from 1.a and be careful about time-dependent parameters.

2.b

Assume that we observe call-option prices (say, for strikes $K = 80, 81, \dots, 120$) coming from a Bachelier model with $r = 0$, $S(0) = 100$, $T = 0.25$, and $\sigma = 15$.

What do the (Black-Scholes) implied volatilities across strikes look like? **Comment** on the results. **What** would happen if we had instead assumed that $S_0 = 50$?

Quanto Hedging and The Kingdom of Denmark Put (50%)

In this exercise we consider an arbitrage-free currency model of Black/Scholes-type. More specifically, we think of and refer to “domestic” as “US” and “foreign” as “Japan”. With Björk’s Proposition 18.7 (with a sign typo corrected) in mind we write bank-accounts, exchange-rate and Japanese stock dynamics under the US-martingale measure as

$$\begin{aligned} dB_{US}(t) &= r_{US}B_{US}(t)dt \\ dX(t) &= X(t)(r_{US} - r_J)dt + X(t)\sigma_X^\top dW(t) \\ dB_J(t) &= r_JB_J(t)dt \\ dS_J(t) &= S_J(t)(r_J - \sigma_X^\top \sigma_J)dt + S_J(t)\sigma_J^\top dW(t), \end{aligned}$$

where the σ ’s are 2-dimensional (constant column) vectors. (We don’t need a US stock here; we’ll get enough fun as it is.)

For numerical experiments, you can/may/should assume that $r_{US} = 0.03$, $r_J = 0.00$, $\sigma_X^\top = (0.1, 0.02)$, and $\sigma_J^\top = (0, 0.25)$.

Consider a *quanto put*. This is an option that at time T pays off

$$Y_0(K - S_J(T))^+,$$

where (the constant) Y_0 is some agreed-upon-in-advance exchange-rate; it could be the time-0 exchange rate or a forward exchange rate.

3.a

Show that the arbitrage-free time- t price of the quanto (aka Kingdom of Denmark – for reasons discussed at the lectures) put is $F^{QP}(t, S_J(t))$, where the function F^{QP} is defined by

$$F^{QP}(t, s) = Y_0 e^{-r_{US}(T-t)} \left(K\Phi(-d_2(t, s)) - e^{(r_J - \sigma_X^\top \sigma_J)(T-t)} s\Phi(-d_1(t, s)) \right),$$

Φ is the standard normal distribution function as usual,

$$d_{1/2}(t, s) = \frac{\ln(s/K) + (r_J - \sigma_X^\top \sigma_J \pm \|\sigma_J\|^2/2)(T-t)}{\sqrt{T-t}\|\sigma_J\|},$$

and $\|\cdot\|$ denotes the Euclidian norm of a vector.

Show that

$$\frac{\partial F^{QP}(t, s)}{\partial s} = Y_0 e^{(r_J - \sigma_X^\top \sigma_J - r_{US})(T-t)} (\Phi(d_1(t, s)) - 1) =: g(t, s).$$

(“=” and “:=” mean “defined as”, with the term closest to “:=” being defined.)

Hint: Careful “pattern recognition” brings you back to Black-Scholes, thus requiring no new calculations.

3.b

Consider now the usual set-up for a discrete hedge experiment, i.e. the interval $[0; T]$ is split into n pieces at the equidistant time-points t_i , where we adjust our portfolio.

Illustrate experimentally (say, in the plausible case where $Y_0 = X(0) = 1/100$ and $K = S_J(0) = 30,000$ and $T = 2$) that a strategy that

- at time t_i holds

$$\Delta^{QP}(t_i, S_J(t_i), X(t_i)) := \frac{g(t_i, S_J(t_i))}{X(t_i)}$$

units of the foreign stock, and

- is then made self-financing with the domestic bank-account,

does not replicate the pay-off of the quanto put (even in the ‘ $n \rightarrow \infty$ ’-limit).

3.c

Suppose now that the strategy from 3.b is amended/extended by holding (at time t_i)

$$-\Delta^{QP}(t_i, S_J(t_i), X(t_i))S_J(t_i)$$

units of currency, which are instantly deposited in the foreign bank (giving you $-\Delta^{QP}S_J/B_J$ units of it), where they earn interest; it is still kept self-financing via domestic bank-account.

Illustrate experimentally that this strategy (“suitably initiated” and “in the limit”) *does* replicate the pay-off of the quanto put.

3.d

Explain why (i.e.: prove mathematically that) the strategy from part 3.c works (replication-wise in continuous time) while the strategy from 3.b does not.

3.e

Assume $dW^{\mathbb{P}} = dW + \varphi(t)dt$, where $W^{\mathbb{P}}$ under the real-world probability measure $\mathbb{P}$ and φ is a Girsanov kernel or risk-premium process.

Suppose X and S_J are $\mathbb{P}$ -martingales. **What** does that imply about the φ -process? **Illustrate experimentally** that the quanto option hedge strategy from 3.c also works when simulations are run under $\mathbb{P}$. **Explain** why.